

ENSCI 99

A Practical Guide to Environmental Choices

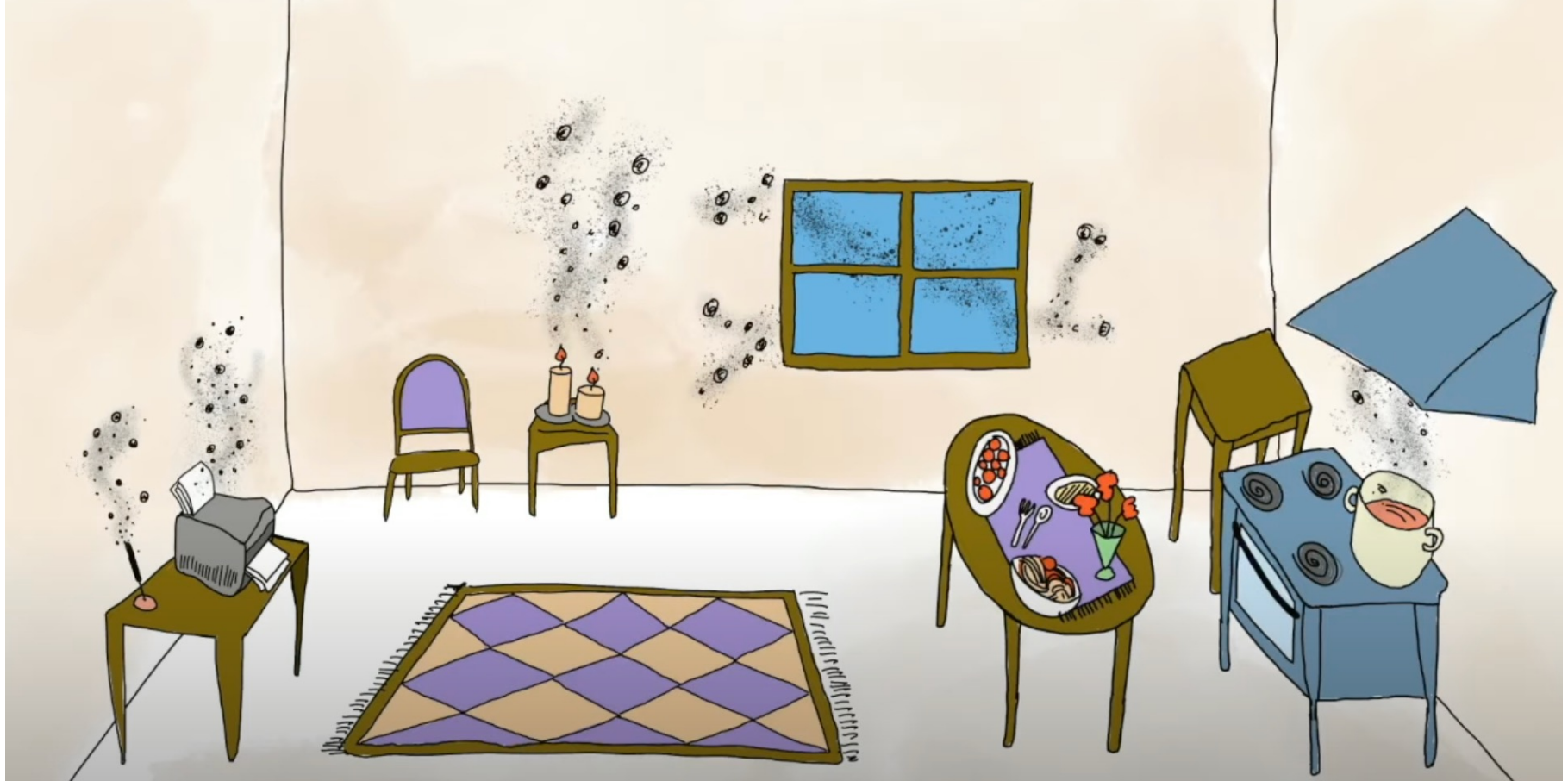
Air Pollution II

Objectives

Air pollution continued

- Indoor air pollution sources
 - Health impacts
- What is being done to control air pollution?
 - The Clean Air Act of 1963
 - Emissions Trading program, 1990
 - Technology
 - Better Choices
 - The Montreal Protocol

Sources of indoor air pollution



https://www.youtube.com/watch?v=_sMuXtjZndY

Air Pollution _2



Pigeon tracks air pollution in London

The Clean Air Act addresses pollution

- Clean Air Act of **1963**
 - Monitoring and air pollution controls
- Amended in
 - **1970**: required regulations for industrial and mobile air pollution sources.
 - **1990**: acid rain prevention program, required permits and enhanced enforcements

National Ambient Air Quality Standards (NAAQS)

Primary: public health, sensitive populations

Secondary: public welfare and environment

- **Six (6) criteria air contaminants**
 - Carbon monoxide (CO),
 - Ground Level Ozone (O₃),
 - Lead (Pb),
 - Nitrogen dioxide (NO₂),
 - Particulate matter 2.5 and 10
 - Sulfur dioxide (SO₂).

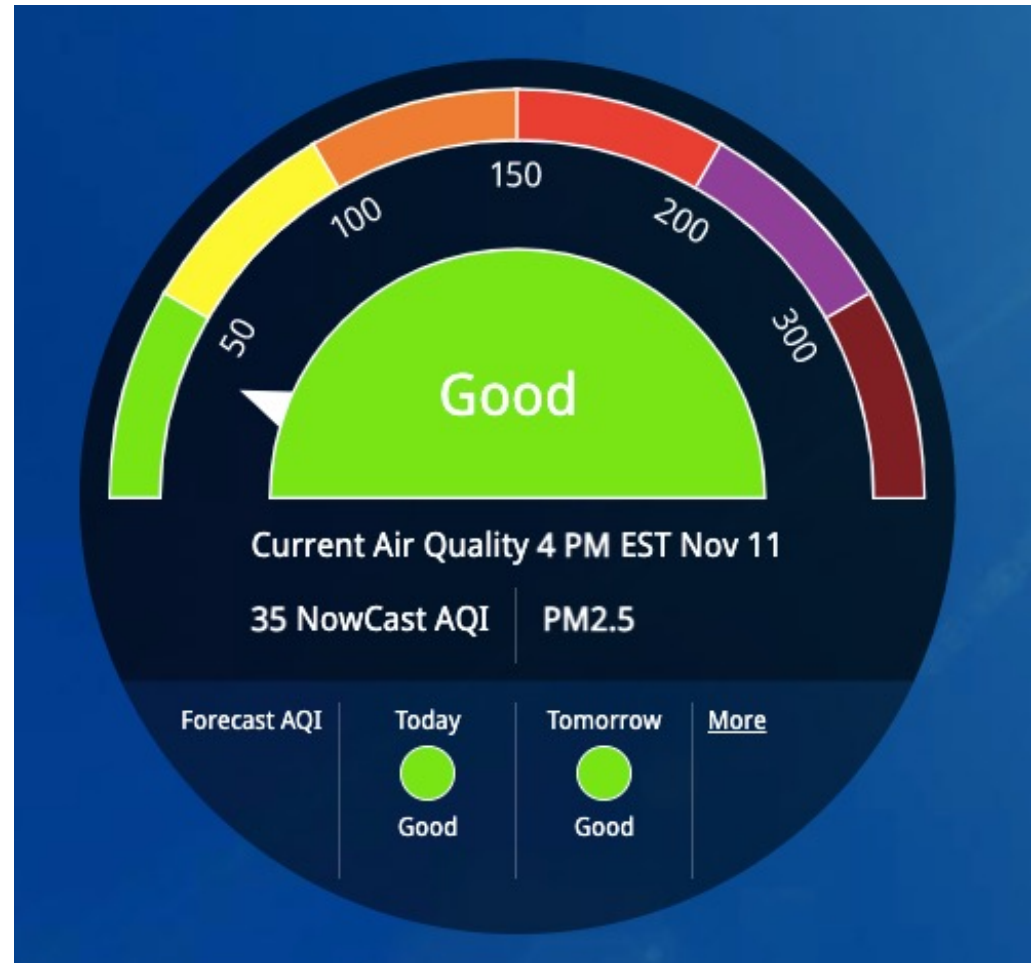
How can we tell if the air outside is safe?

AQI – Air Quality Index

- Indicates whether pollutant levels in air may cause health concerns.
- Ranges from 0 (least concern) to 500 (greatest concern).

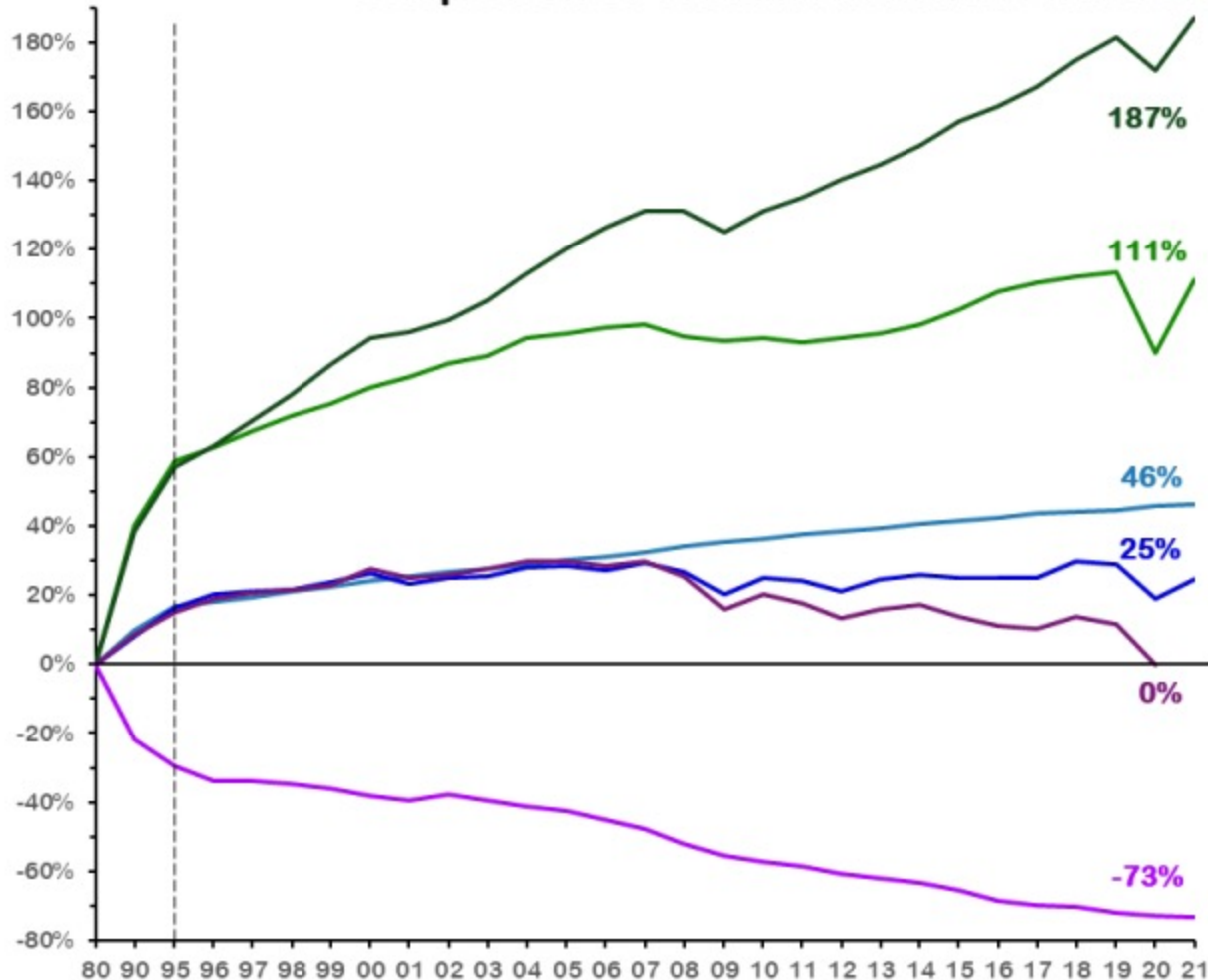
Government agencies

- For current air quality, visit [AirNow.gov](https://airnow.gov).



How has the Clean Air Act affected air pollution trends?

Comparison of Growth Areas and Emissions, 1980-2021



Gross Domestic Product



Vehicles Miles Traveled



Population



Energy Consumption



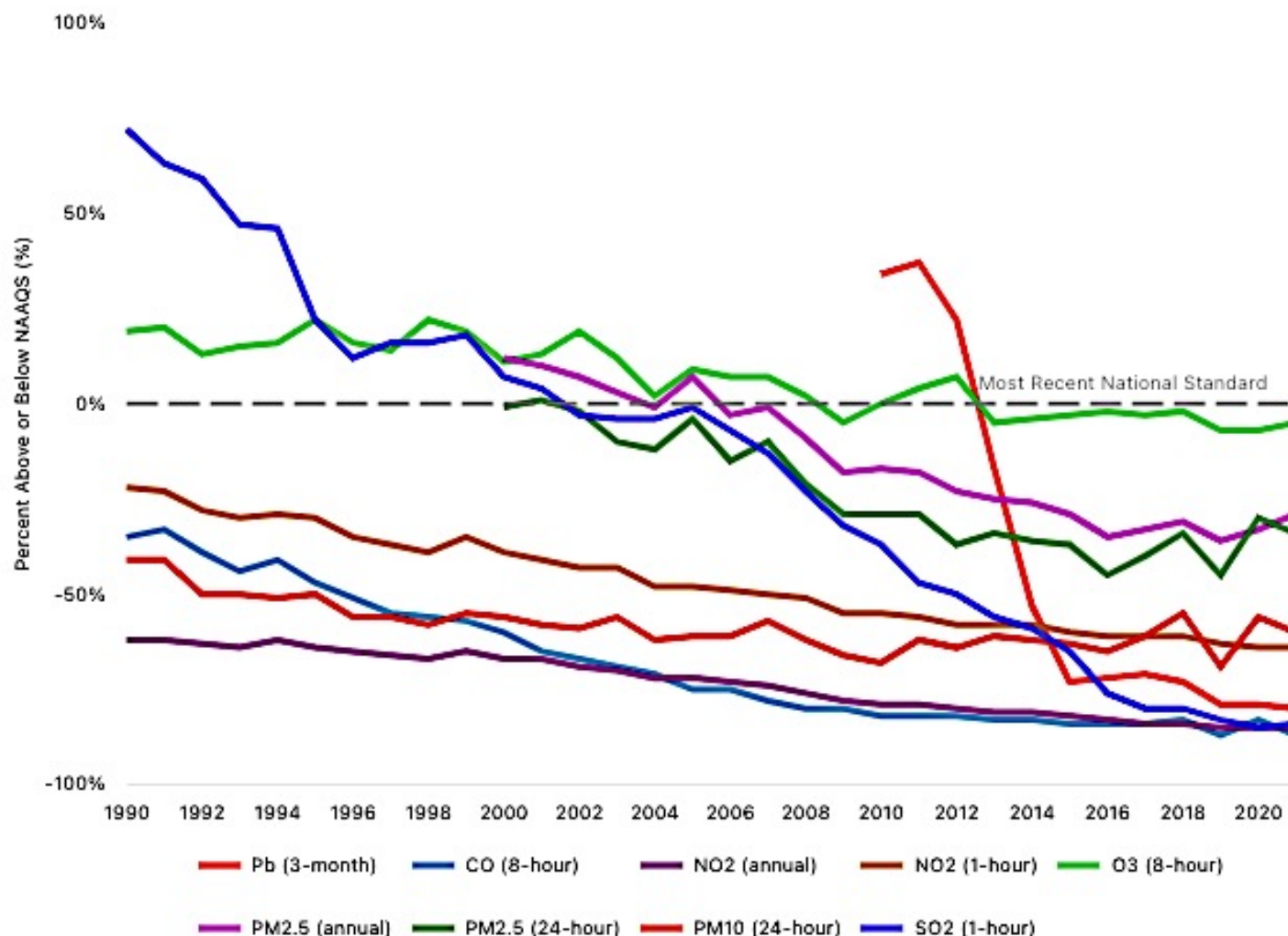
CO₂ Emissions



Aggregate Emissions
(Six Common Pollutants)

Concentration of criteria pollutants in ambient air across the US has steadily fallen

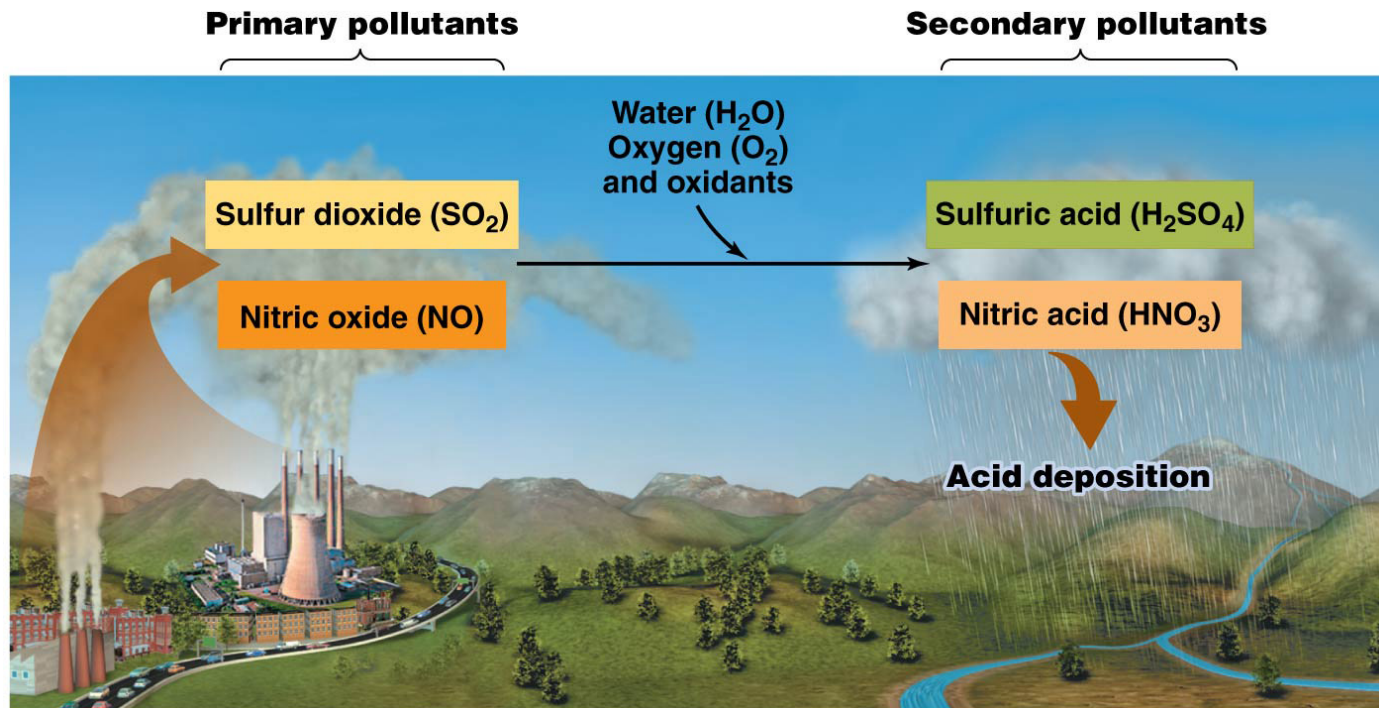
Declining National Air Pollutant Concentration Averages



Source: U.S. EPA Air Quality System

Burning fossil fuels produces acid rain

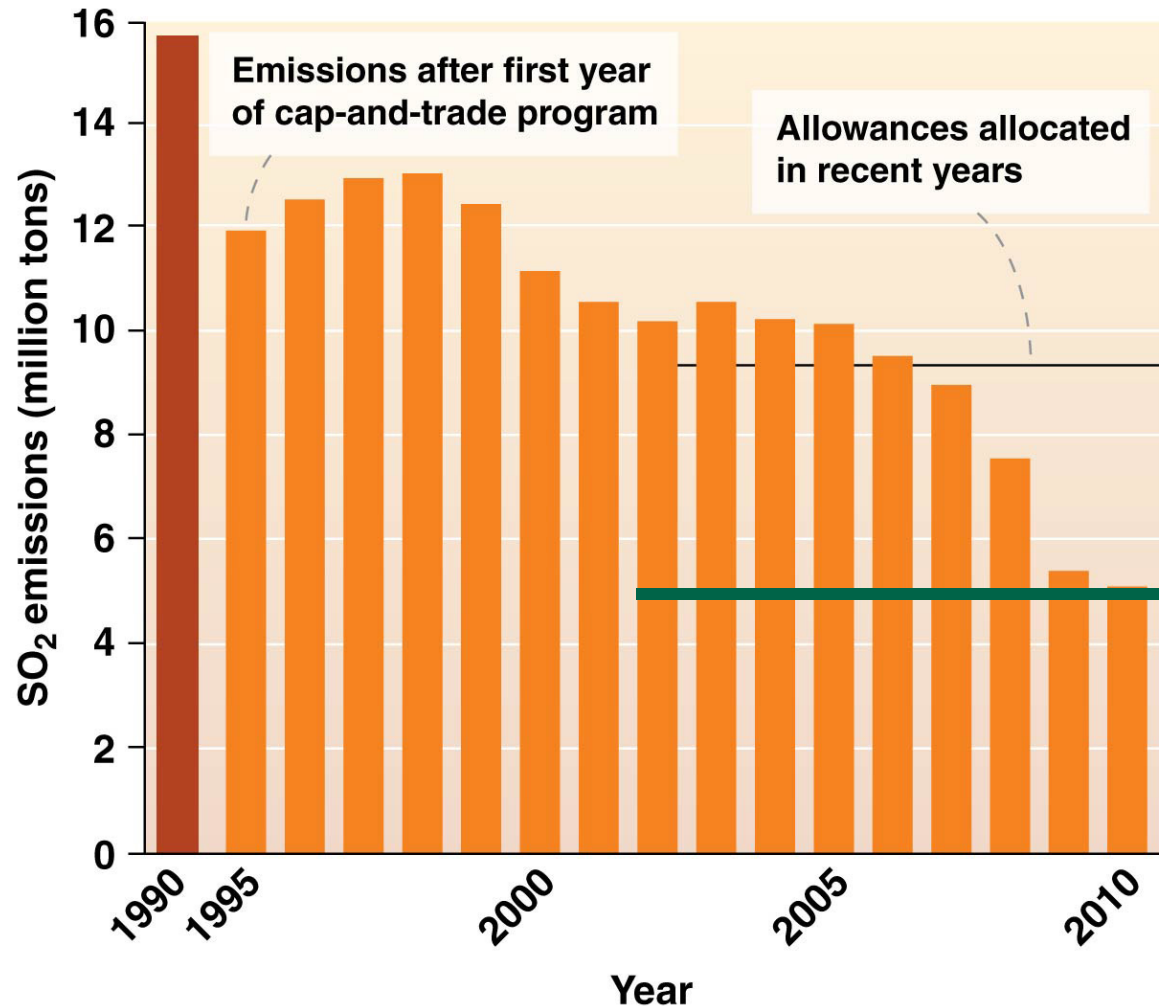
- Burning fossil fuels releases sulfur dioxide and nitrogen oxides
 - These compounds react with water, oxygen, and oxidants to form sulfuric and nitric acids



How did we address the acid rain problem?

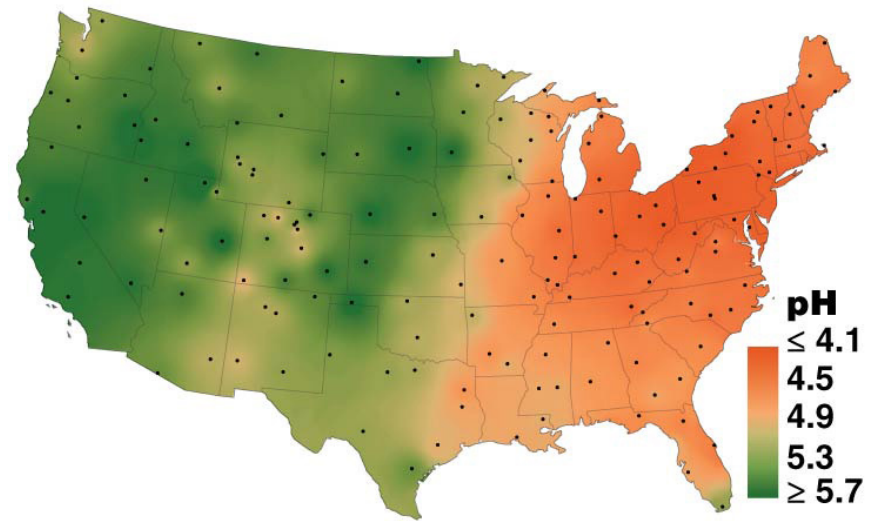
- The Clear Air Act (1990) established an **emissions trading program** for sulfur dioxide
- Low sulfur coal
- New technologies such as scrubbers have helped
- Policies and regulations have lowered U.S. emissions of sulfur and nitrogen oxides 67% since 1989

How did the **emissions trading system** impact sulfur dioxide emissions?

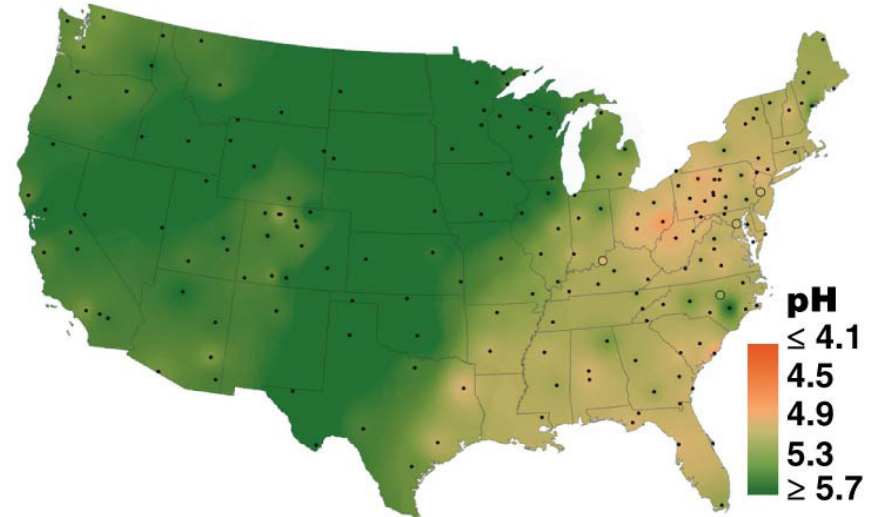


Sulfur dioxide emissions fell 67% since the emissions trading system

- How has the pH of precipitation changed under the Clean Air Act?
- Precipitation has become less acidic as air quality has improved under the Clean Air Act

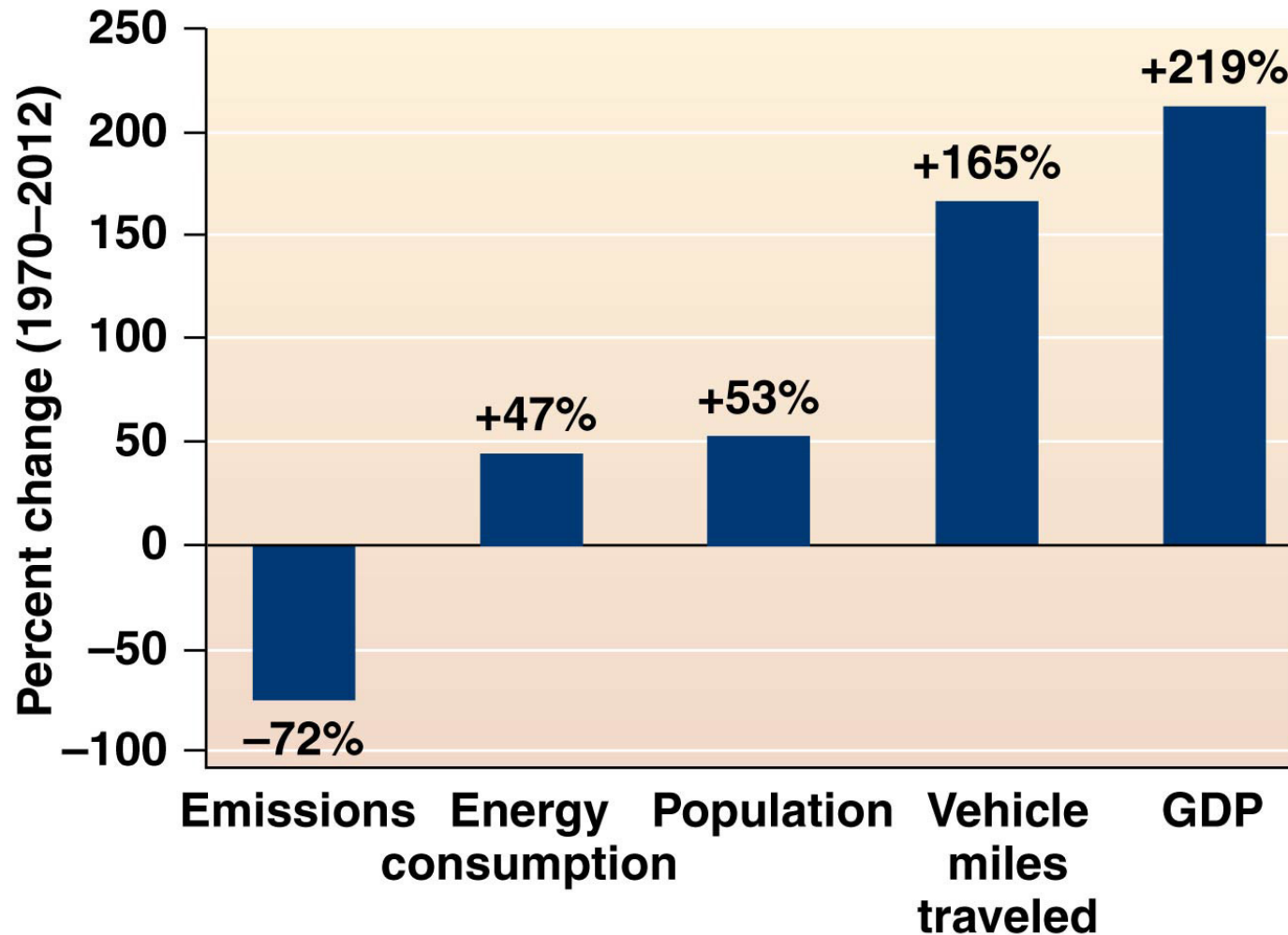


(a) pH of precipitation in 1990



(b) pH of precipitation in 2012

How has the Clean Air Act affected growth?



(b) Trends in major indicators

How was it possible to reduce pollutant emissions?

- Federal policies and technology
 - Cleaner burning fuels
 - Fuel efficiency
- Reduced SO₂ emissions
 - Permit trading programs and clean coal technologies



How was it possible to reduce pollutant emissions?

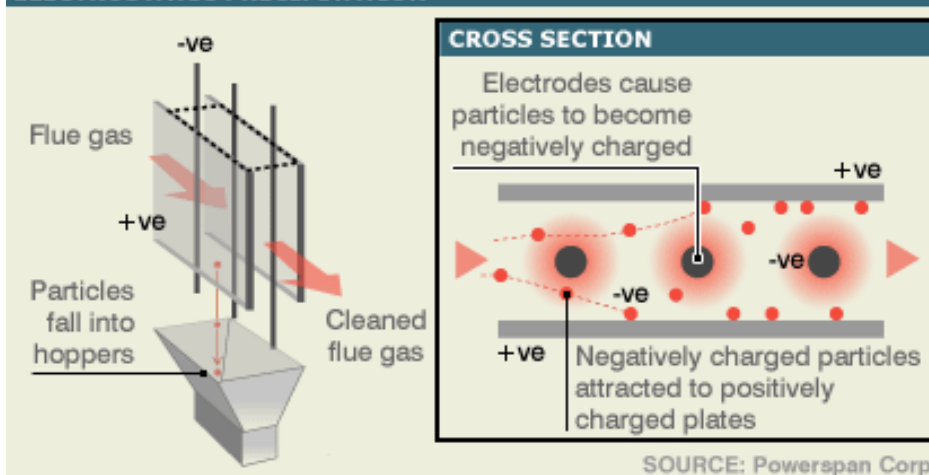
- Scrubbers
 - Chemically convert or physically remove pollutants before they leave smokestacks
- Two main types
 - Wet – liquids
 - Dry - filters



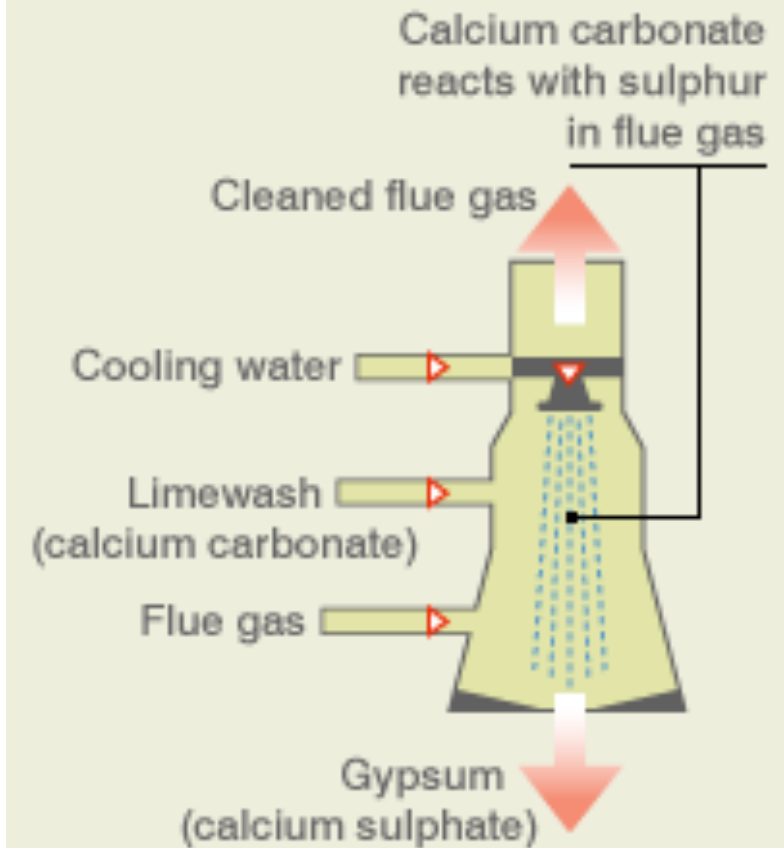
How was it possible to reduce pollutant emissions?

Burning coal produces a range of pollutants that harm the environment: Sulphur dioxide (acid rain); nitrogen oxides (ground-level ozone) and particulates (affects people's respiratory systems)

ELECTROSTATIC PRECIPITATION



REMOVING SULPHUR DIOXIDE



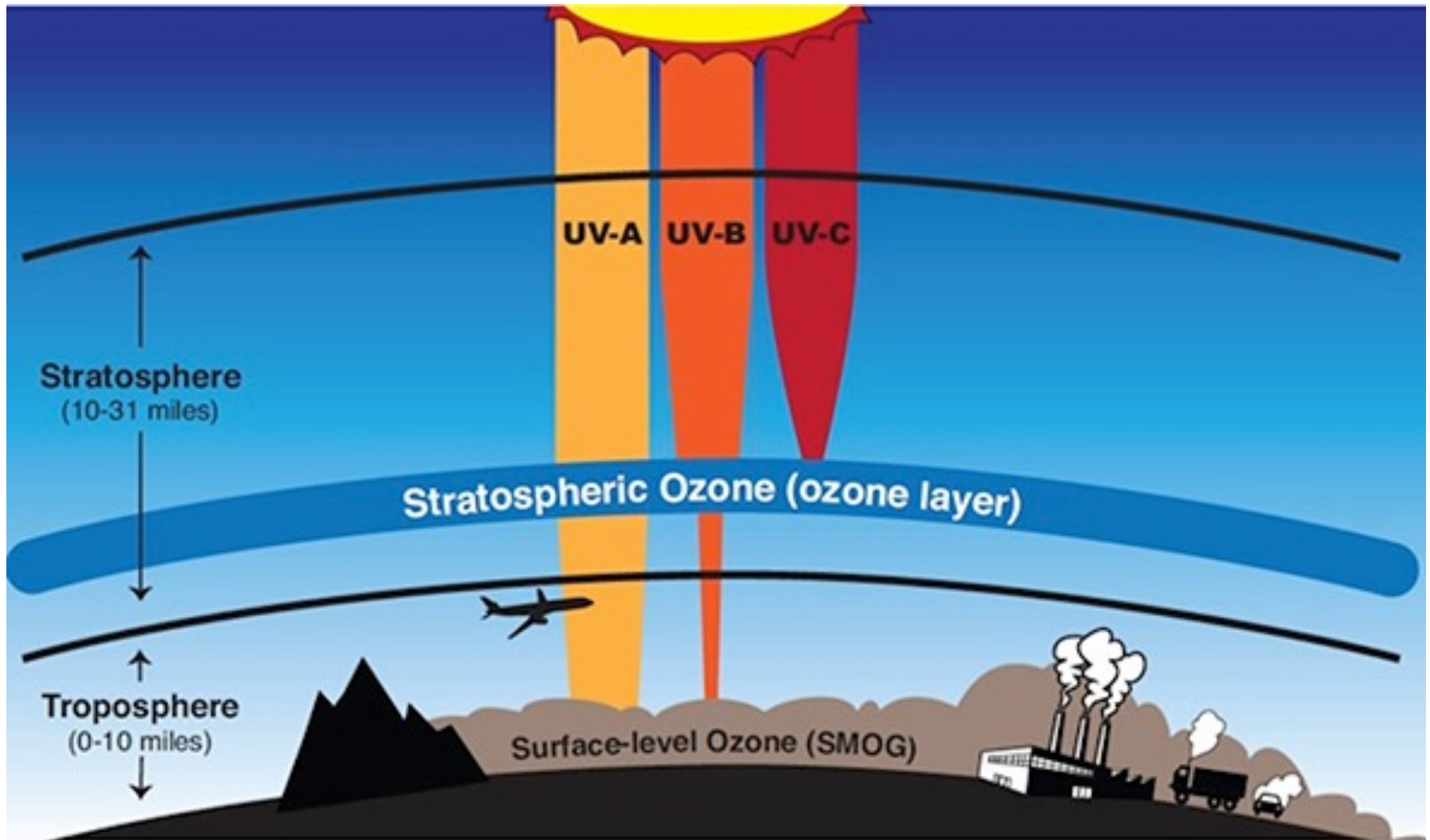
What are some choices we can make to improve air quality?

- **Energy efficient vehicles**
 - Electric vehicles, proper vehicle maintenance
- **Utilizing public transportation**
 - Reduce per person emissions
- **Active transportation**
 - Walking, cycling
- **Eco-friendly products**
 - Low VOC paints, cleaning products
- **Supporting sustainable agriculture**
 - Minimize emissions

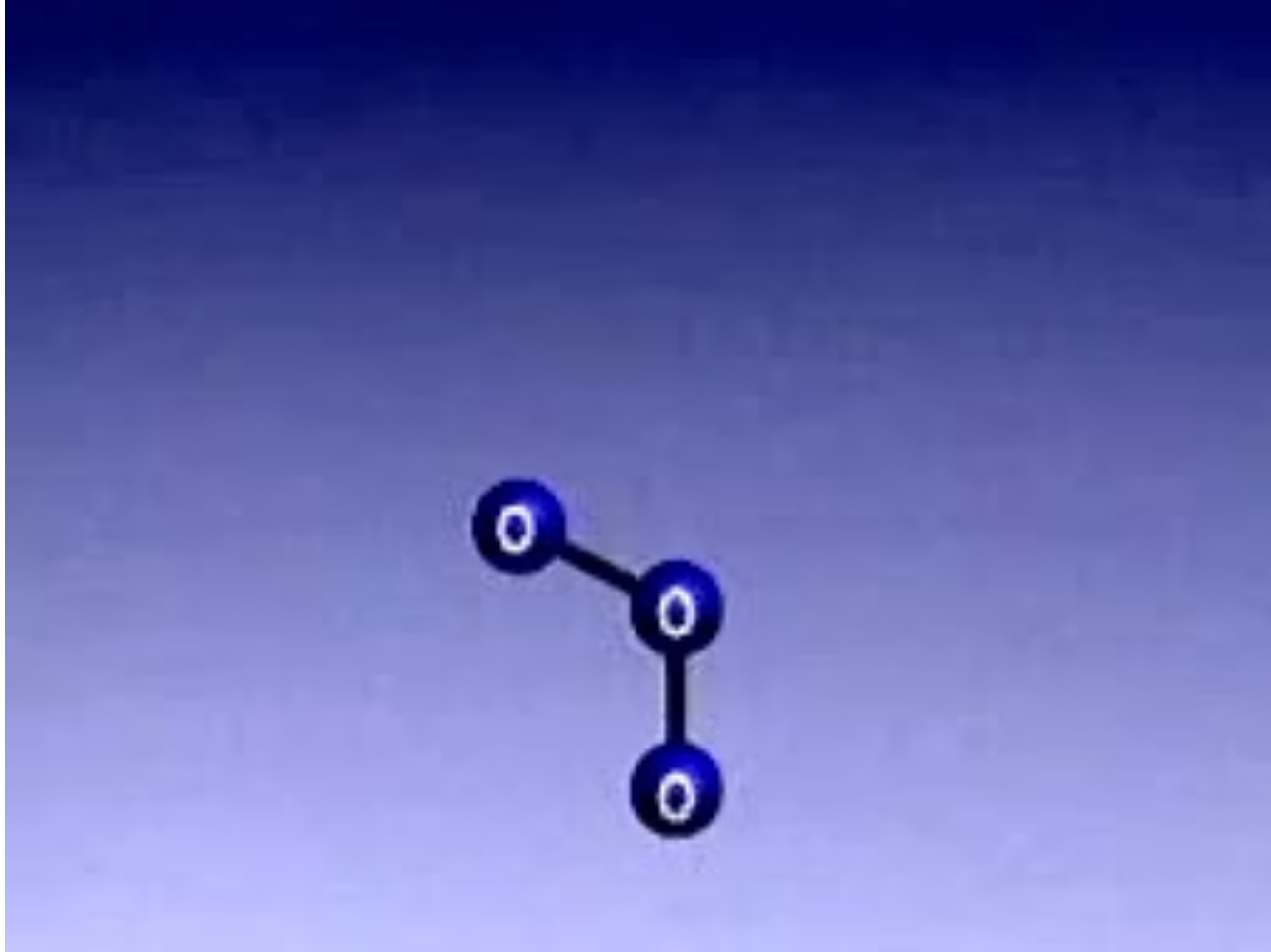
What is the ozone layer?

- Ozone in the lower stratosphere (**good ozone**) absorbs the sun's ultraviolet (UV) radiation
 - UV radiation can damage tissues and DNA
- **Ozone-depleting substances**, chlorofluorocarbons (CFCs)
 - Human-made chemicals that destroy ozone
- **Halocarbons**
 - Human-made compounds made from hydrocarbons with added chlorine, bromine, or fluorine

What is the ozone layer?



How do Chlorofluorocarbons (CFCs) work to destroy ozone?

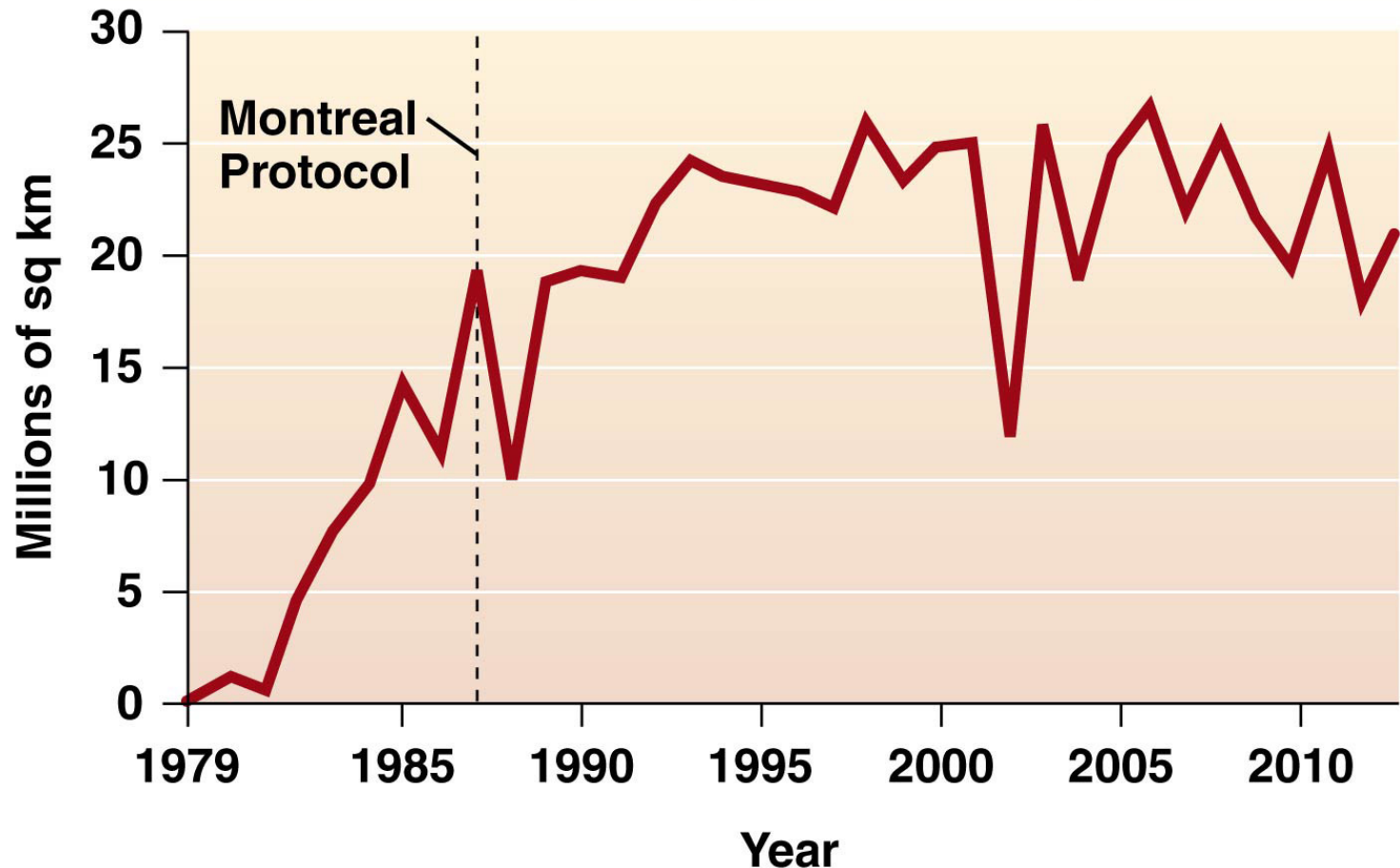


How did we address ozone depletion?

- **Montreal Protocol (1987)**
 - 196 nations agreed to cut CFC production in half by 1998
 - Later agreements deepened cuts, advanced timetables, and addressed other ozone-depleting chemicals
- Industry shifted to safer alternative chemicals
- We stopped the Antarctic ozone hole from getting worse
- **Challenges still face us**
 - CFCs will remain in the stratosphere for decades

Do you think the Montreal treaty worked?

- Phasing out ozone-depleting substances in 1987 worked—the Antarctic ozone hole stopped growing



Let's Talk

Groups: 3-4 people (Include all names)



How are technological innovations like scrubbers, shaping our approach to controlling air pollution? What are possible key challenges & opportunities?

Possible challenges

Cost

- **Installing and maintenance**

Energy Consumption

- **Adds to a facility's overall energy footprint**

Waste management

- **Wastewater generated need special treatment & disposal**

Limited effectiveness

- **Efficiency varies depending of type of pollutant**

Opportunities

Improved public health

- **Better air quality improves public health and quality of life**

Environmental benefits

- **Compliance with air quality regulations, decrease overall environmental impact**

Technological advancements

- **Develop more efficient and cost effective technologies**

Market potential

- **Growing demand for cleaner air creates opportunities for manufacturing and innovation**